• Examen Recuperatorio 1

_ Ejercicio 1

a) $r^2 = x_i x_i$

$f(r)$ función escalar arbitraria dependiente de r

$$\left.\begin{aligned} &\frac{\partial}{\partial x_j}(r^2) = 2r \cdot \frac{\partial r}{\partial x_j} \\ &\frac{\partial}{\partial x_j}(x_i x_i) = \underbrace{\frac{\partial x_i}{\partial x_j}}_{\delta_{ij}} \cdot x_i + x_i \cdot \frac{\partial x_i}{\partial x_j} = 2x_j \end{aligned}\right\} \Rightarrow \frac{\partial r}{\partial x_j} = \frac{\cancel{2} x_j}{\cancel{2} r} = \frac{x_j}{r}$$

$$\left[\underline{\nabla} f(r)\right]_j = \frac{\partial f(r)}{\partial x_j} = \frac{\partial f(r)}{\partial r} \cdot \frac{\partial r}{\partial x_j} = \frac{\partial f(r)}{\partial r} \cdot \frac{x_j}{r} \Rightarrow$$

$$\Rightarrow \underline{\nabla} f(r) = \frac{\underline{x}}{r} \cdot \frac{\partial f(r)}{\partial r} \qquad \text{con} \quad r = \|\underline{x}\|$$

b)

$$\left[\underline{\nabla}(\underline{x} \cdot \underline{\underline{A}}\,\underline{x})\right]_i = \frac{\partial}{\partial x_i}(x_m A_{mn} x_n) \quad \wedge \quad A \text{ constante en el dominio} \Rightarrow$$

$$\frac{\partial}{\partial x_i}(x_m A_{mn} x_n) = \underbrace{\frac{\partial x_m}{\partial x_i}}_{\delta_{mi}} \cdot A_{mn} \cdot x_n + x_m \cdot A_{mn} \cdot \underbrace{\frac{\partial x_n}{\partial x_i}}_{\delta_{ni}} = A_{in} \cdot x_n + x_m \cdot A_{mi} =$$

$$= A_{in} \cdot x_n + A^T_{im} x_m = A_{im} x_m + A^T_{im} x_m = (A_{im} + A^T_{im}) x_m \Rightarrow$$

$$\Rightarrow \underline{\nabla}(\underline{x} \cdot \underline{\underline{A}}\,\underline{x}) = (\underline{\underline{A}} + \underline{\underline{A}}^T)\underline{x}$$

c)

Asumiendo dim = 3

$$\varepsilon_{jmn} \varepsilon_{imn} = \varepsilon_{njm} \varepsilon_{nim} = \delta_{ji} \overbrace{\delta_{mm}}^{3} - \delta_{jm} \delta_{mi} = 3\delta_{ji} - \delta_{ji} = 2\underbrace{\delta_{ji}}_{\delta_{ij} \text{ por simetría}} = 2\delta_{ij}$$

- Ejercicio 2

$$A = A_{ijk}\, x_i\, x_j\, x_k \qquad \wedge \quad A_{ijk} \text{ constantes en el dominio}$$

$$\frac{\partial A}{\partial x_i} = \frac{\partial}{\partial x_i}\left(A_{ljk}\, x_l\, x_j\, x_k\right) = A_{ljk}\left(\underbrace{\frac{\partial x_l}{\partial x_i}}_{\delta_{li}} \cdot x_j \cdot x_k + x_l \cdot \underbrace{\frac{\partial x_j}{\partial x_i}}_{\delta_{ji}} \cdot x_k + x_l \cdot x_j \cdot \underbrace{\frac{\partial x_k}{\partial x_i}}_{\delta_{ki}}\right)$$

$$= A_{ijk}\, x_j x_k + A_{lik}\, x_l x_k + A_{lji}\, x_l x_j =$$

índices dummy

$$= A_{ijk}\, x_j x_k + A_{jik}\, x_j x_k + A_{kji}\, x_k x_j =$$

$$= \left(A_{ijk} + A_{jik} + A_{kji}\right) x_j x_k$$

$$\frac{\partial^2 A}{\partial x_i\, \partial x_j} = \frac{\partial}{\partial x_j}\left[\left(A_{ilk} + A_{lik} + A_{kli}\right) \cdot x_l x_k\right] =$$

$$= \left(A_{ilk} + A_{lik} + A_{kli}\right) \cdot \left(\underbrace{\frac{\partial x_l}{\partial x_j}}_{\delta_{lj}} \cdot x_k + x_l \cdot \underbrace{\frac{\partial x_k}{\partial x_j}}_{\delta_{kj}}\right) =$$

$$= \left(A_{ijk} + A_{jik} + A_{kji}\right) x_k + \left(A_{ilj} + A_{lij} + A_{jli}\right) x_l =$$

Por índice dummy

$$= \left(A_{ijk} + A_{jik} + A_{kji}\right) x_k + \left(A_{ikj} + A_{kij} + A_{jki}\right) x_k =$$

$$= \left(A_{ijk} + A_{jik} + A_{kji} + A_{kij} + A_{ikj} + A_{jki}\right) x_k$$

Ejercicio 3

a) $\frac{\partial \sigma_{ij}}{\partial x_j} = -b_i$

$$\frac{\partial \sigma_{11}}{\partial x_1} + \frac{\partial \sigma_{12}}{\partial x_2} + \frac{\partial \sigma_{13}}{\partial x_3} = b_1$$

$$\cancel{-2x_1x_2} + \cancel{2x_2x_1} + 0 = b_1 \Rightarrow b_1 = 0$$

$$\frac{\partial \sigma_{21}}{\partial x_1} + \frac{\partial \sigma_{22}}{\partial x_2} + \frac{\partial \sigma_{23}}{\partial x_3} = b_2$$

$$\cancel{-(4-x_2^2)} - \frac{1}{\cancel{3}}\cancel{(\overset{1}{\cancel{3}}x_2^2 - \overset{4}{\cancel{12}})} + 0 = b_2 \Rightarrow b_2 = 0$$

$$\frac{\partial \sigma_{31}}{\partial x_1} + \frac{\partial \sigma_{32}}{\partial x_2} + \frac{\partial \sigma_{33}}{\partial x_3} = b_3$$

$$0 + 0 + 0 = b_3 \Rightarrow b_3 = 0$$

b)

Evaluación del tensor de tensiones en $P(2,-1,6)$

$\sigma_{xx} = (1-4)(-1) = 3$

$\sigma_{yy} = -\frac{1}{3}[-1 - 12(-1)] = -11/3$

$\sigma_{xy} = -(4-1) \cdot 2 = -6$

$\sigma_{zz} = (3-4)(-1) = 1$

$$\underline{\underline{\Upsilon}} = \begin{bmatrix} 3 & -6 & 0 \\ -6 & -11/3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Autovalores:

$$\left| \Upsilon_{ij} - \lambda^{(i)} \delta_{ij} \right| = 0$$

$$\begin{vmatrix} 3-\lambda & -6 & 0 \\ -6 & -11/3-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (-1)^{3+3} (1-\lambda) \cdot \begin{vmatrix} 3-\lambda & -6 \\ -6 & -\frac{11}{3}-\lambda \end{vmatrix} =$$

$$= \cancel{(-1)^{3+3}} (1-\lambda) \left[(3-\lambda) \cdot \left(-\frac{11}{3} - \lambda\right) - 36 \right] = 0 \Rightarrow$$

$$\Rightarrow \lambda = 1 \quad \wedge \quad -11 - 3\lambda + \frac{11}{3}\lambda + \lambda^2 - 36 = 0$$

$$\lambda^2 + \frac{2}{3}\lambda - 47 = 0 \quad \times 3$$

$$3\lambda^2 + 2\lambda - 141 = 0$$

$$\lambda = \frac{-2 \pm \sqrt{4 - 4.3(-141)}}{2.3} = \frac{-2 \pm \sqrt{1696}}{6} = -\frac{1}{3} \pm \frac{\sqrt{1696}}{6}$$

$$= -\frac{1}{3} \pm \frac{\sqrt{4^2 . 106}}{6} = -\frac{1}{3} \pm \frac{2}{3}\sqrt{106} = \frac{1}{3}\left(-1 \pm 2\sqrt{106}\right)$$

$$\begin{cases} \lambda = \frac{1}{3}\left(-1 + 2\sqrt{106}\right) \approx 6{,}530 \\ \lambda = \frac{1}{3}\left(-1 - 2\sqrt{106}\right) \approx -7{,}197 \end{cases}$$

$\lambda_1 = 6{,}530 = \Upsilon_{máx}$ //

$\lambda_2 = 1$ //

$\lambda_3 = -7{,}197 = \Upsilon_{mín}$ //

c)

Corte máximo:

$$\tau_{máx} = \frac{\sigma_{max} - \sigma_{min}}{2} = \frac{\frac{1}{3}\left[\cancel{-1} + \cancel{2}\sqrt{106} - \left(\cancel{-1} - \cancel{2}\sqrt{106}\right)\right]}{\cancel{2}} = \frac{2}{3}\sqrt{106} \approx 6{,}864$$

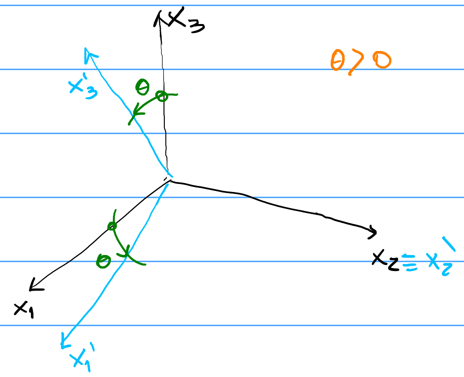
d)

Plano $x_1 + 3x_2 + x_3 = 1 \Rightarrow \underline{n}^* = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$

$$\underline{n} = \frac{1}{\sqrt{1^2 + 3^2 + 1^2}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{11}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$

$$\underline{T} = \underline{\underline{\sigma}} \cdot \underline{n} = \begin{bmatrix} 3 & -6 & 0 \\ -6 & -11/3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \frac{1}{\sqrt{11}} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{11}} \cdot \begin{bmatrix} -15 \\ -17 \\ 1 \end{bmatrix}$$

- Ejercicio 4



$$\beta_{ij} = \underline{e}_i' \cdot \underline{e}_j = \cos(\widehat{\underline{e}_i' \underline{e}_j})$$

$$\beta_{22} = \cos(0) = 1, \quad \beta_{32} = \beta_{23} = \beta_{12} = \beta_{21} = \cos\left(\frac{\pi}{2}\right) = 0$$

$$\beta_{11} = \beta_{33} = \cos\theta$$

$$\beta_{13} = \cos\left(\theta + \frac{\pi}{2}\right) = -\operatorname{sen}\theta$$

$$\beta_{31} = \cos\left(\frac{\pi}{2} - \theta\right) = \operatorname{sen}\theta$$

$$\underline{\underline{\beta}} = \begin{bmatrix} \cos\theta & 0 & -\operatorname{sen}\theta \\ 0 & 1 & 0 \\ \operatorname{sen}\theta & 0 & \cos\theta \end{bmatrix}$$

$$\Gamma_{ij}' = \beta_{im}\beta_{jn}\Gamma_{mn} = \beta_{im}\Gamma_{mn}\beta_{nj}^T \Rightarrow \underline{\underline{\Gamma}}' = \underline{\underline{\beta}}\,\underline{\underline{\Gamma}}\,\underline{\underline{\beta}}^T$$

$$\begin{bmatrix} \tau & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & \tau \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & \operatorname{sen}\theta \\ 0 & 1 & 0 \\ -\operatorname{sen}\theta & 0 & \cos\theta \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & 0 & -\operatorname{sen}\theta \\ 0 & 1 & 0 \\ \operatorname{sen}\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \tau\cos\theta & 0 & -\tau\operatorname{sen}\theta \\ 0 & \alpha & 0 \\ \tau\operatorname{sen}\theta & 0 & \tau\cos\theta \end{bmatrix} \begin{bmatrix} \tau\overbrace{\cos^2\theta + \tau\operatorname{sen}^2\theta}^{1} & 0 & \tau\cancel{\cos\theta}\cdot\operatorname{sen}\theta - \tau\cancel{\operatorname{sen}\theta}\cos\theta \\ 0 & \alpha & 0 \\ \tau\operatorname{sen}\theta\cancel{\cos\theta} - \tau\cos\theta\cancel{\operatorname{sen}\theta} & 0 & \tau\cdot\overbrace{\operatorname{sen}^2\theta + \tau\cos^2\theta}^{1} \end{bmatrix}$$

$$\underline{\underline{\Gamma}}' = \begin{bmatrix} \tau & 0 & 0 \\ 0 & \alpha & 0 \\ 0 & 0 & \tau \end{bmatrix} = \underline{\underline{\Gamma}} \qquad \forall\theta$$

El tensor dado es invariante para rotaciones alrededor del eje x_2. Eso se condice que en el plano x_1-x_3, donde se realiza la rotación, las componentes $\underline{\underline{\Upsilon}}$ tienen características de un tensor isotrópico en 2D:

$$\underline{\underline{\Upsilon}}^{(x_1-x_2)} = \tau \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \tau \cdot \underline{\underline{I}}^{(x_1-x_2)}$$

y la única componente de $\underline{\underline{\Upsilon}}$ en x_2 no nula es Υ_{22} (que es lo que ocurre si x_1, x_2 y x_3 son ejes ppales.)

Solución alternativa:

Se divide el tensor $\underline{\underline{\Upsilon}}$ de la siguiente forma:

$$\underline{\underline{\Upsilon}} = \tau \cdot \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\underline{\underline{I}}-\underline{\underline{A}}} + \alpha \cdot \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}}_{\underline{\underline{A}}} = \tau\left(\underline{\underline{I}} - \underline{\underline{A}}\right) + \alpha \cdot \underline{\underline{A}}$$

$$\underline{\underline{B}} = \begin{bmatrix} \cos\theta & 0 & -\operatorname{sen}\theta \\ 0 & 1 & 0 \\ \operatorname{sen}\theta & 0 & \cos\theta \end{bmatrix}$$

rotación alrededor del eje x_2

Nota que: $\underline{\underline{B}} \cdot \underline{\underline{A}} = \underline{\underline{A}}$

y que trasponiendo $\underbrace{\underline{\underline{A}}^T}_{\underline{\underline{A}}} \cdot \underline{\underline{B}}^T = \underbrace{\underline{\underline{A}}^T}_{\underline{\underline{A}}} \Rightarrow \underline{\underline{A}} \cdot \underline{\underline{B}}^T = \underline{\underline{A}}$

$$\underline{\underline{B}} \cdot \underline{\underline{\Upsilon}} = \tau \underline{\underline{B}}\left(\underline{\underline{I}} - \underline{\underline{A}}\right) + \alpha \underline{\underline{B}} \cdot \underline{\underline{A}} = \tau\left(\underline{\underline{B}} - \underline{\underline{A}}\right) + \alpha \underline{\underline{A}}$$

$$\underline{\underline{\Upsilon}}' = \underline{\underline{B}} \cdot \underline{\underline{\Upsilon}} \cdot \underline{\underline{B}}^T = \tau\left(\overbrace{\underline{\underline{B}} \cdot \underline{\underline{B}}^T}^{\underline{\underline{I}}} - \underline{\underline{A}}\,\underline{\underline{B}}^T\right) + \alpha\, \underline{\underline{A}}\,\underline{\underline{B}}^T = \tau\left(\underline{\underline{I}} - \underline{\underline{A}}\right) + \alpha \underline{\underline{A}} = \underline{\underline{\Upsilon}} \quad \forall\theta$$

- Ejercicio 5

Ver notas de clase.